

Dhimitrios Duka

Graduate Research Assistant
M.Sc. Computer Science, Saarland University



EDUCATION

Saarland University

10/2023 – Present

Master of Science: Computer Science, Current GPA: 1.6 (best 1.0, worst 4.0).
Thesis: Temporal Boundary Refinement for Improved Egocentric Vision-Language Pre-training
Advisors: Prof. Dr. Bernt Schiele and Dr. Anna Kukleva

Polytechnic University of Tirana

10/2018 – 03/2022

Bachelor of Science: Computer Engineering, GPA: 9.7/10.0 (best 10.0, worst 5.0).
Thesis: A Neural Image Caption Generator for the Albanian language
Advisors: Prof. Dr. Elinda Kajo Meçe & Dr. Evis Trandafili

PUBLICATIONS

- [1] Sjarhei Sheludsko, **Dhimitrios Duka**, Bernt Schiele, Hilde Kuehne, Anna Kukleva. 2026. MM-TS: Multi-Modal Temperature and Margin Schedules for Contrastive Learning with Long-Tail Data. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*. [Oral Presentation at WACV 2026]
- [2] **Dhimitrios Duka**, Bernt Schiele, Hilde Kuehne, Anna Kukleva. 2026. TeMo: Temperature Modulation for Multimodal Contrastive Learning. *Under Submission*.

TALKS

- [1] Invited talk: "MM-TS: Multi-Modal Temperature and Margin Schedules for Contrastive Learning with Long-Tail Data," Department of Computer Engineering, Middle East Technical University.
- [2] Seminar talk on **Generative Models in Computer Vision (GMCV)**, Seminar led by Dr. Jan Eric Lenssen, 08/2025.
- [3] Seminar talk on **Explainable Machine Learning (ExML)**, Seminar led by Dr. Jonas Fischer, 08/2024.

RESEARCH EXPERIENCE

Max Planck Institute for Informatics

Saarland, Germany

Research Assistant

11/2024 – Present

Advisors: Dr. Anna Kukleva, Prof. Dr. Bernt Schiele and Prof. Dr. Hilde Kuehne

- [1] Researched supervised fine-tuning (SFT) and Group Relative Policy Optimization (GRPO) for temporal action localization and refinement in large-scale egocentric videos.
- [2] Co-authored MM-TS (WACV 2026 Oral) and TeMo (Under Submission), developing novel temperature scaling methods that achieve state-of-the-art performance on long-tail multimodal benchmarks.
- [3] Designed and executed large-scale deep learning experiments on a multi-GPU SLURM cluster, building robust pipelines for multimodal representation learning.

INDUSTRY EXPERIENCE

Sciant

Tirana, Albania

Full-Stack Software Engineer

08/2023 – 10/2023

- [1] Built a backend integration service for Protel using Spring Boot, Node.js, and AWS (S3, DynamoDB), enabling reliable and scalable data synchronization.
- [2] Developed comprehensive documentation and automated tests to improve system reliability, maintainability, and long-term operational support.

DigitSapient

Tirana, Albania

Full-Stack Software Engineer

08/2021 – 07/2023

- [1] Built secure, end-to-end digital workflows for the Unified Patent Court e-Justice platform, meeting strict reliability, privacy, and compliance requirements.
- [2] Modernized the platform by migrating from a monolithic architecture to PostgreSQL-backed microservices, enabling greater scalability and easier service integration.

AWARDS

1st Place, Computer Graphics Rendering Competition, Saarland University

02/2024

SELECTED PROJECTS

sCode – Open-Source VS Code Extension

2025 – Present

- [1] Built an open-source VS Code extension for managing SLURM clusters, enabling job submission, monitoring, and resource inspection directly from the editor.
- [2] Developed interactive dashboards and cluster management tools using TypeScript and the VS Code Extension API to streamline HPC workflows.

Applied LLM Evaluation & Fine-Tuning

03/2024

- [1] Designed a multilingual LLM evaluation framework for XGLM-564M and GPT-2 on FLORES, analyzing cross-lingual generation and representation quality.
- [2] Applied PEFT methods (BitFit, LoRA, IA3) for low-resource language adaptation, reducing training loss by 30.4% while updating fewer than 0.28% of model parameters.

Neural Image Captioning for Albanian

09/2021 – 03/2022

- [1] Built an end-to-end vision-language pipeline for Albanian by creating a caption dataset through pretrained neural machine translation.
- [2] Trained an encoder-decoder image captioning model achieving 12.4 BLEU-4 and improved caption quality through decoding strategy optimization.

TECHNICAL SKILLS

AI & Foundation Models: Large Language Models (LLMs), Generative AI, Supervised Fine-Tuning (SFT), Reinforcement Learning (GRPO), PEFT (LoRA, BitFit, IA3), Contrastive Learning, Prompt Engineering, Vision-Language Pipelines

Frameworks & Compute: PyTorch, Hugging Face Transformers, vLLM, Weights & Biases, SLURM, Multi-GPU environments, Docker, Kubernetes, AWS

Software Engineering: Python, TypeScript, Java, Node.js, React, PostgreSQL, CI/CD, Git

Languages: English (Full Professional Proficiency), German (A1), Albanian (Native)